

IK 542 : Machine Learning for Sanskrit Text Analysis

Course Code : IK 542

Course Name : Upanishads and Vedanta Studies

L-T-P-C : 3-0-0-3

Intended for : BTech/MTech/MS/MSc/MA/Ph.D

Prerequisite : Basic Computer Fundamentals

Mutual Exclusion: None

Approval: 42nd Senate

Course Contents

- **Vyakaran:** Phonetics in Ancient India, W S Allen 1971, Sandhi, W S Wallen, Morphology, Syntax, Karaka Analysis, Indian Theories of Meaning, Philosophy of Word and Meaning, Sanskrit Philosophy of Language, Logic, Language, Reality. The Sanskrit Language: An Overview - History and Structure, Linguistic and Philosophical Representations, Annotation schemes for sandhi, morph, compound, karaka, An introduction to sanskrit language and linguistics. (12 Hours)
- **Natural Language Processing – 1:** Overview of NLP and its applications, Historical perspective and evolution of NLP, Challenges in natural language understanding and generation, Basic linguistic concepts (syntax, semantics, pragmatics), Text Processing and Preprocessing, Language Models and Probability in NLP. (12 hours)
- **Machine Learning-1(Introductory):** Definition of Machine Learning, Applications of Machine Learning, Overview of the Machine Learning Process, Machine Learning for NLP, Project Work and Presentations on NLP. (16 hours)

Textbooks:

1. Akshar Bharati, Vineet Chaitanya, Rajeev Sangal, **NLP: A Pāṇinian perspective**, prentice hall of India, 1995
2. Daniel Jurafsky and James H Martin, **Speech and Language Processing**.
3. **Explorations in Artificial Intelligence and Machine Learning**, A CRC Press FreeBook, Taylor and Francis Group
4. Dr. Naresh Jha, **Ashtadhyayi of Pāṇini**, Chaukhamba Surbharati Prakashan, 2014
5. Amba Kulkarni, **Sanskrit Parsing: Based on the Theories of Shabdabodha**, D K Printworld and IIAS Shimla, 2019.
6. Kapil Kapoor, **Dimensions of Pāṇini Grammar**, D K Print World, 2020

References:

1. Bloomfield, **Language**, Motilal Banarsidass.
2. Prof. Korada Subramaniam, **Theories of Language: Oriental and Occidental**.



3. Institute Franciasde Pondicherry, RS Vidyapeeth, 2006
4. M Sriman Narayan Murti, **An introduction to Sanskrit Linguistics**, DK Publications.
5. Mark Lutz, **Programming Python: Powerful Object-Oriented Programming**, 4th Edition, O'Reilly, 2011.